

DINGLASAN, KARL AKIM C.

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EDUCATION

Pamantasan ng Lungsod ng San Pablo

2024 – 2028

Bachelor of Science in Computer Engineering

San Pablo City, Laguna

- Relevant Coursework: Programming Logic and Design, Data Structures and Algorithms, Fundamentals of Electronics Circuits, Database Systems, Computer Networks
- Activities: Technical Committee Head, Computer Engineering Student Society

EXPERIENCE

Technical Committee Head

2025 – Present

Computer Engineering Student Society

Pamantasan ng Lungsod ng San Pablo

- Lead a 5-member Technical Committee, overseeing technical initiatives, events, and internal tools for the organization.
- Plan and organize technical workshops, seminars, and competitions for fellow Computer Engineering students.
- Manage the organization's technical infrastructure, including its website and social media presence.
- Coordinate with committee members to delegate tasks and ensure timely delivery of technical projects and events.

PROJECTS

IoT Water Monitor — Smart Water Distribution Prototype

A.Y. 2025–2026

Team Capstone Project (in progress)

- Building a working prototype of a smart water monitoring system using real ESP32 hardware and sensors (flow, pressure, leak) to model how the system would function in an actual school building.
- Stack: Node.js/Express, MQTT, InfluxDB, MariaDB (Dockerized); React + TypeScript dashboard; companion Flutter app with Telegram bot alerts.
- Built the Flutter app's UI/theming and contributed to backend data pipeline debugging, alongside the development team. Designed around targets including $\geq 85\%$ leak detection accuracy and ≤ 15 -second shutoff response.

PLSP Lost and Found Management System

A.Y. 2025

Team Project (3 members)

- Built a full-stack web app digitizing the university's lost-and-found process — report, search, and claim lost/found items. Stack: Next.js, Go (Gin), PostgreSQL, Docker.
- Contributed across most of the stack, from API logic to frontend implementation.

12V Regulated DC Power Supply — Circuit Design & PCB Fabrication

May 2026

Team Course Project (6 members) — AC222: Fundamentals of Electronics Circuits

- Designed, fabricated, and tested a regulated power supply converting 220V AC into a stable 12V DC output: simulated in Proteus, PCB designed in EasyEDA, fabricated via manual etching, then soldered and assembled.
- Verified a stable 12V DC output under load using a multimeter and LED string load test.

SKILLS

- **Languages & Frameworks:** React, Next.js, TypeScript, Node.js, Flutter, Python, Java, Go, C/C++ (Embedded)
- **Tools & Platforms:** Docker, MQTT, InfluxDB, SQL (PostgreSQL/MySQL), Git, Proteus, EasyEDA
- **Hardware:** ESP32, Circuit Design, PCB Design, IoT Sensors
- **Other:** Leadership, Team Collaboration, Project Management
- **Languages:** Filipino (Native), English (Proficient)

SEMINARS & TRAININGS

Master Agile Project Success — PRINCE2® Agile® Foundation & Practitioner Training

Jan 2025

KOENIG Webinar (via Zoom)

VMware NSX — Network Virtualization & Security in Hybrid Clouds

Jan 2025

KOENIG Webinar (via Zoom)

Python's Role in AI-Driven Automation

Feb 2025

KOENIG Webinar (via Zoom)